# Learning in the Age of AI

## Lecture 02: Human Learning Psychology

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18 March 2026

# Memory Is Not Exposure

## Guiding Question

Why does rereading often feel productive while retrieval practice produces stronger long-term learning?

$$\text{retrieval} > \text{re-exposure}$$

Speaker notes

Explain the distinction between familiarity and retrievability — students often confuse the two.

# Why Might This Be True?

- rereading creates fluency
- fluency feels like understanding
- retrieval forces reconstruction

# Interpretation

If you cannot bring the idea back without support, the idea is not stable yet ([?BrownRoedigerMcDaniel2014]; [?Dunlosky2013]).

Test-enhanced learning shows retrieval improves delayed retention, even when restudying looks better immediately after ([?RoedigerKarpicke2006]; [?KarpickeBlunt2011]).

# Evidence

- **Immediate impression:** rereading creates familiarity
- **Delayed outcome:** retrieval produces stronger later recall
- **Didactic implication:** ask students to explain from memory before showing the polished answer

> The right test of learning is not recognition now, but reconstruction later.

# Forgetting and Spacing

## Guiding Question

Why can forgetting actually help learning?

$$R = e^{-t/s}$$

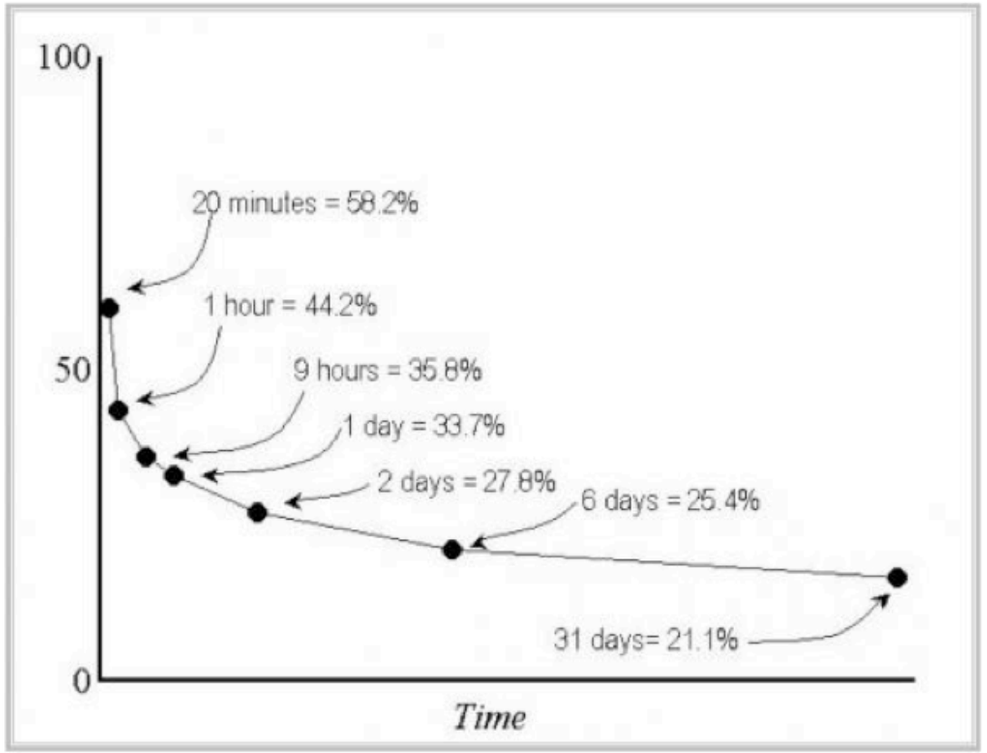
# Why Might This Be True?

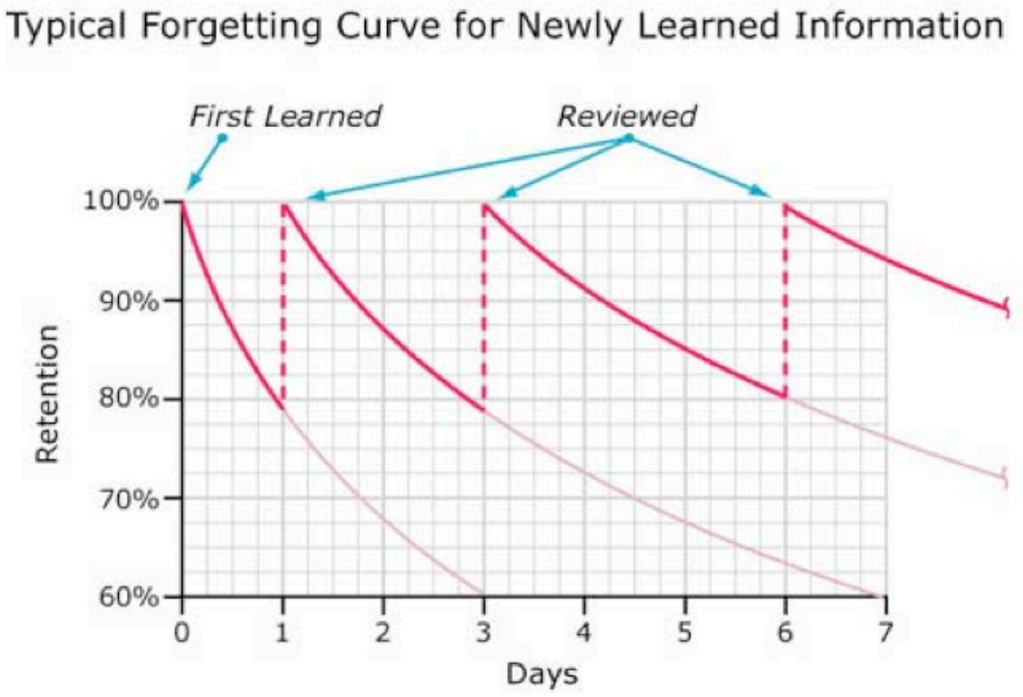
- memory weakens over time
- effortful retrieval strengthens access
- spacing makes retrieval effortful again

# Interpretation

Spacing works not despite difficulty but because of it ([?Ebbinghaus1885]; [?BjorkBjork2011]).

# Evidence




- Ebbinghaus established the basic empirical pattern: retention drops sharply without reinforcement ([?Ebbinghaus1885])
- distributed practice outperforms massed practice across many recall tasks ([?Cepeda2006])
- the practical point is *not* "wait as long as possible" but "revisit after enough delay that retrieval requires effort"

Repeat after some forgetting, not only before forgetting starts.

# Doing Beats Hearing

## Guiding Question

Why do interaction and explanation usually outperform listening alone?

$$\text{interactive} > \text{constructive} > \text{active} > \text{passive}$$

# Why Might This Be True?

- passive listening creates little transformation
- constructive activity creates inferences
- interaction exposes misunderstandings

# Interpretation

Discussion matters because it forces students to externalize, compare, revise, and defend understanding ([?Chi2009]).

# Evidence

- **Passive:** hearing or reading without transformation
- **Active:** marking or selecting
- **Constructive:** generating explanations or examples
- **Interactive:** reasoning with another person and adapting to their response

> Durable learning requires doing, explaining, and interacting, not only hearing.

# Studying With AI

## Guiding Question

How should students use AI without weakening their own learning?

$$\text{tool support} + \text{retrieval} + \text{self-explanation} + \text{discussion}$$

# Why Might This Be True?

- AI improves access to examples and reformulations
- AI can also create fluency illusions
- only retrieval and explanation reveal whether understanding is real

# Interpretation

Use AI to generate contrasts, examples, and counterpositions. Then explain the concept *without* the tool and test whether the idea still holds together.

# Evidence

- ask AI for alternative explanations, not final authority
- close the tool and reconstruct the idea from memory
- compare your answer with the AI output and identify what you missed
- discuss the mismatch with peers or in class

> Use AI as a sparring partner, not as a replacement for interpretation and retrieval.